

Explicit Local Integrals of Motion for the Many-Body Localized State

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Recently, it has been suggested that the many-body localized phase can be characterized by local integrals of motion. Here we introduce a Hilbert-space-preserving renormalization scheme that iteratively finds such integrals of motion exactly. Our method is based on the consecutive action of a similarity transformation using displacement operators. We show, as a proof of principle, localization and the delocalization transition in interacting fermion chains with random on-site potentials. Our scheme of consecutive displacement transformations can be used to study many-body localization in any dimension, as well as disorder-free Hamiltonians.

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Since the revival of interest in localization due to disorder [1–4], it has been suggested that the so-called many-body localized (MBL) phase can be characterized by an extensive set of local integrals of motion (LIOMs) or l bits, τ_i^z , that commute with each other and the Hamiltonian [5–10]. Consequently, the Hamiltonian can be written in terms of these LIOMs as

$$H = \sum_i \xi_i \tau_i^z + \sum_{ij} V_{ij} \tau_i^z \tau_j^z + \dots \quad (1)$$

Many properties of the MBL phase, such as its logarithmic entanglement spread or its insulating behavior, can be derived based on this assumption [10].

The questions that remain, however, are what those LIOMs are and how to compute them. Since any sum and product of integrals of motion is itself an integral of motion, the choice of LIOMs is highly arbitrary. Purely mathematically, all projectors onto the (localized) eigenstates are integrals of motion, and out of those one could in principle construct the local integrals of motion. In fact, it is easy to show that all Hamiltonians can be brought into the form dictated by Eq. (1) [11]. As for the MBL phase, Chandran *et al.* [5] use the long-time-evolved average of an initially local operator as their LIOMs, whereas Ros *et al.* [9] and Imbrie [7] use perturbative methods to construct local integrals of motion.

In this Letter, we construct iteratively a transformation that turns *any* fermionic Hamiltonian into the classical form of Eq. (1). This is done by consecutively applying a similarity transformation using a displacement operator $\exp[\lambda(X^\dagger - X)]$. The elegant properties of this transformation allow for a systematic elimination of off-diagonal interaction terms, order by order in the number of fermionic operators involved. Our renormalization scheme can be used to study Hamiltonians in any dimension, with or without disorder.

Whether a random interacting system is localized or not depends on how much the integrals of motion τ_i^z are spread out. As a proof of principle, we apply our method to diagonalize random interacting chains. In both the localized and the delocalized regime we find agreement with exact diagonalization (ED). Throughout the phase diagram we find the effective interactions between the integrals of motion, and we can infer the exponential localization of the integrals of motion in the localized regime.

Definitions.—Here we will consider interacting fermions with random on-site potentials [2,3]

$$H = \sum_\alpha \xi_\alpha c_\alpha^\dagger c_\alpha + \frac{1}{2} \sum_{\alpha\beta\gamma\delta} V_{\alpha\beta\gamma\delta} c_\alpha^\dagger c_\beta^\dagger c_\gamma c_\delta, \quad (2)$$

as it extends the original concept of Anderson localization [1] to interacting systems. For this model we will now characterize all possible terms in the Hamiltonian.

We define a classical term in the Hamiltonian as a product of fermionic density operators of the form $n_{i_1} n_{i_2} \dots n_{i_d}$ where all i 's are different. A quantum term is the product of fermionic operators that cannot be written as a classical term. It thus contains, next to a possible set of density operators, separate creation and annihilation operators,

$$X = n_{i_1} n_{i_2} \dots n_{i_d} c_{j_1}^\dagger c_{j_2} c_{j_3}^\dagger c_{j_4} \dots c_{j_q}. \quad (3)$$

Again, all the i 's and all the j 's are different from each other. The order $\mathcal{O}(X)$ of a term is defined as the total number of creation and annihilation operators it contains. This equals $\mathcal{O}(X) = 2d + q$, where d is the number of density operators and q is the number of separate creation and annihilation operators. We require the order to be even, and the interaction to contain the same number of creation and annihilation operators, so that all interactions preserve the total fermion number [12]. We call the notation of Eq. (3)

normal ordered: all density interactions are grouped together, and the remaining is a product of alternating creation and annihilation operators that all act on different sites.

The order of a sum of terms is defined as the minimum of the orders of the individual terms,

$$\mathcal{O}\left(\sum_i X_i\right) = \min_i [\mathcal{O}(X_i)]. \quad (4)$$

When multiplying two terms X and Y , the product XY can contain interaction terms of order lower than $\mathcal{O}(X) + \mathcal{O}(Y)$. This happens when X contains the annihilation operator on site α and Y contains the creation operator on the same site; we call this an overlap. This gives $c_\alpha c_\alpha^\dagger = 1 - n_\alpha$, which generates a term that has an order that is two lower. As this can happen for any pair of creation and annihilation operators in X and Y , there exists a case of maximal overlap with overlap on $\frac{1}{2}\min[\mathcal{O}(X), \mathcal{O}(Y)]$ sites. Because each such overlap generates a term with an order two lower, the product XY has order

$$\mathcal{O}(X) + \mathcal{O}(Y) \geq \mathcal{O}(XY) \geq \max[\mathcal{O}(X), \mathcal{O}(Y)]. \quad (5)$$

There are three important properties of quantum terms: (i) The product of a quantum term with itself is zero, $X^\dagger X^\dagger = XX = 0$. (ii) The product of a quantum term with its Hermitian conjugate, i.e., $X^\dagger X$ and $XX^\dagger$, is classical. Observe that because X and $X^\dagger$ have maximal overlap, the order remains the same: $\mathcal{O}(XX^\dagger) = \mathcal{O}(X^\dagger X) = \mathcal{O}(X)$. (iii) The cubic power is trivial, that is, $X^\dagger XX^\dagger = X^\dagger$ and $XX^\dagger X = X$.

Displacement transformation.—Based on the classification of terms we just introduced, we can define a displacement operator associated with a quantum term X ,

$$\mathcal{D}_X(\lambda) = \exp[\lambda(X^\dagger - X)]. \quad (6)$$

Because of the aforementioned properties of quantum terms, the displacement operator can be written out explicitly as

$$\mathcal{D}_X(\lambda) = 1 + \sin \lambda (X^\dagger - X) + (\cos \lambda - 1)(X^\dagger X + XX^\dagger). \quad (7)$$

Note that the Hermitian conjugate of the displacement operator is $\mathcal{D}_X^\dagger(\lambda) = \mathcal{D}_X(-\lambda)$. The above arguments can easily be extended to spin- $\frac{1}{2}$ Hamiltonians, where classical terms are given by products of S^z operators, and quantum terms are total spin-conserving products of S^+ , S^- , and S^z operators.

The displacement transformation is given by a similarity transformation using the displacement operator. That is, it transforms any term Y as

$$Y \rightarrow \tilde{Y} = \mathcal{D}_X^\dagger(\lambda) Y \mathcal{D}_X(\lambda). \quad (8)$$

This transformation is similar to a Clifford group rotation, which has transformation operator $\mathcal{D} = e^{\lambda A}$ with $A^2 = 1$; here, we have the weaker condition $A^3 = -A$.

For now we use the notation with the tilde to denote the transformed term; we will drop the tilde later as we will perform many consecutive transformations. Under the transformation, there are “new” terms *generated*, namely, $\tilde{Y} - Y$. Using the explicit formulation of the displacement operator Eq. (7), we see that the new terms are of the form XY , YX , etc. Carefully counting all the combinations, we see that the order of the new terms is at least the maximum of the orders of X and Y ,

$$\mathcal{O}(\tilde{Y} - Y) \geq \max[\mathcal{O}(X), \mathcal{O}(Y)]. \quad (9)$$

As will be shown later, this lower bound on the order of new terms implies the closedness of our systematic transformation procedure.

Without constraining the specific shape of X that we use for the displacement transformation, we can prove that the only way to generate new terms proportional to $X^\dagger + X$ is through terms that have maximal overlap with X [13]. For example, consider an order-4 term $X^\dagger = c_1^\dagger c_2 c_3^\dagger c_4$ with interaction given by V . Let us write down the relevant part of the Hamiltonian as follows:

$$\sum_{i=1}^4 \xi_i n_i + V_{13} n_1 n_3 + V_{24} n_2 n_4 + \frac{1}{2} V (X^\dagger + X). \quad (10)$$

The displacement transformation with $\mathcal{D}_X(\lambda)$ leaves the quadratic part untouched, and the prefactor multiplying $(X^\dagger + X)$ becomes

$$\frac{1}{2} V \cos 2\lambda + \frac{1}{2} (\xi_1 + \xi_3 + V_{13} - \xi_2 - \xi_4 - V_{24}) \sin 2\lambda, \quad (11)$$

so that, with λ given by

$$\tan 2\lambda = -\frac{V}{\xi_1 + \xi_3 + V_{13} - \xi_2 - \xi_4 - V_{24}}, \quad (12)$$

the transformed Hamiltonian no longer has the interaction term $X^\dagger + X$. Similar expressions can be found for transformations involving X of higher order.

The right-hand side of Eq. (12) equals the “small” parameter that is used in perturbative studies of MBL [2,3,7]. Such perturbation theories often run into the problem of resonances, where the denominator of the small parameter goes to zero, which means that perturbation theory cannot be applied. However, the displacement transformation we present here is well behaved at a resonance, because then $\lambda = \pi/4$ and the interaction can

be still transformed away. Note that each displacement transformation can be viewed as a discrete version of a Wegner transformation [14].

Consecutive displacement transformations.—Now, any fermionic Hamiltonian that respects the total fermion number conservation can be written as

$$H = \sum_i \xi_i n_i + \sum_{n=4,6,\dots} \sum_j V_{nj} (X_{nj}^\dagger + X_{nj}), \quad (13)$$

where ξ_i are the on-site energies, n expresses the order of the term X_{nj} , j is just an index, and V_{nj} are the coupling constants. If all terms X are classical, we have reached our goal: we have a classical Hamiltonian with an infinite set of conserved quantities.

Any quantum term ($X^\dagger + X$) can be removed from the Hamiltonian by performing a displacement transformation associated with X , using the value of λ given by Eq. (12). After doing so, we can choose another quantum term and transform that one away, continuing this path of consecutive transformations.

New terms that are generated are multiplied by either $\sin(\lambda)$, $\cos \lambda$, or products of those. Therefore, generically, new terms have smaller couplings constants, making the process of consecutive transformations to be like a renormalization scheme. This generation of new terms takes into account cotunneling and, hence, long-distance resonances. Note that our scheme preserves the Hilbert space, since each displacement transformation does not decimate sites or bonds, or any other degree of freedom.

In certain cases, however, transforming a term X away can generate terms with an even larger coupling constant. This does not pose a problem: whenever we transform a term X away and it is later regenerated, upon regeneration it will have a smaller coupling constant than before. Additionally, throughout consecutive transformations the distribution of coupling constants will change such that creating larger coupling constants will become more and more unlikely. Therefore, the magnitude of the strongest coupling constant decreases exponentially with the number of applied transformations [13].

We thus remove, term by term, all quantum terms of order 4 in the Hamiltonian. The price we pay is the generation of new terms (both classical and quantum) of order 6 and higher, and new classical terms of order 4. As a result, we obtain a complicated Hamiltonian that is classical in its quadratic and quartic terms. Subsequently, we can do the same tricks for the next-order quantum terms, making the Hamiltonian at that level classical as well, and so forth. The procedure cutoff at n th order reproduces the exact spectrum for states with $n/2$ particles or fewer. The computational complexity of diagonalizing a Hamiltonian up to n th order is $\mathcal{O}(N^{3n/2})$, which we confirmed numerically [13].

In practice, one needs to cut off the procedure at a certain order n . This approximation has a clear physical

interpretation: we are expressing many-body states systematically in terms of n -particle states.

Numerical implementation.—As a proof of principle, we implemented our method numerically. We consider an open chain of L sites with spinless fermions, with a random on-site energy ϵ_i on each site chosen uniformly between $-W/2$ and $W/2$, hopping $t = 1$, and a nearest neighbor repulsion with $V = 1$,

$$H = \sum_{i=1}^N \epsilon_i n_i + t \sum_{i=1}^{N-1} (c_i^\dagger c_{i+1} + c_{i+1}^\dagger c_i) + V \sum_{i=1}^{N-1} n_i n_{i+1}. \quad (14)$$

The Hamiltonian is first diagonalized at the quadratic level; after that, we continue with displacement transformations at quartic order.

At each iteration we pick the quantum term with the largest coupling constant at a given order and transform it away. We neglect coupling constants smaller than numerical accuracy, set at $\epsilon = 3 \times 10^{-2}$. The Hamiltonian is thus diagonalized order by order, up to eighth order, yielding

$$H = \xi_i \tau_i^z + V_{ij} \tau_i^z \tau_j^z + V_{ijk} \tau_i^z \tau_j^z \tau_k^z + V_{ijkl} \tau_i^z \tau_j^z \tau_k^z \tau_l^z. \quad (15)$$

The results are averaged over 20 disorder realizations for $L = 24$ length chains, and 30 realizations at $L = 20$.

Deep in the disordered phase, for $W = 8$, the coefficients in the Hamiltonian fall off exponentially with the maximum distance of the sites involved. This is quantified by computing the average of the absolute value of the coefficients, $\langle |V_{i_1, \dots, i_n}| \rangle$, as shown in Fig. 1. Note that the strength of the coupling constants appears to be independent of the number of density terms involved.

One advantage of our method is that we can also compute the classical Hamiltonian in the delocalized phase. In Fig. 2 we show the distance dependence of the Hamiltonian coefficients as a function of disorder, ranging from $W = 9$ to $W = 3$. If the disorder is weaker than $W_c \approx 5$ the coefficients become independent of distance, which signals delocalization.

To test the accuracy of our method, we compared the ground-state energy obtained using our method with exact diagonalization results for $L = 14$; see the inset of Fig. 2. With only terms up to fourth order, we get more than 95% accuracy in the localized phase. However, close to the transition and in the delocalized regime, higher-order terms are necessary to approach the ED results. For the delocalized regime, the eighth order terms seem to be sufficient, yet at the critical point we need even more-body interactions. This suggests our method becomes very expensive close to the localization-delocalization transition.

Having thus established that the practical implementation of our scheme correctly reproduces ED results in the localized regime, we can directly probe the locality of the integrals of motion. The integrals of motion are given by transforming the initial density operators n_i ,

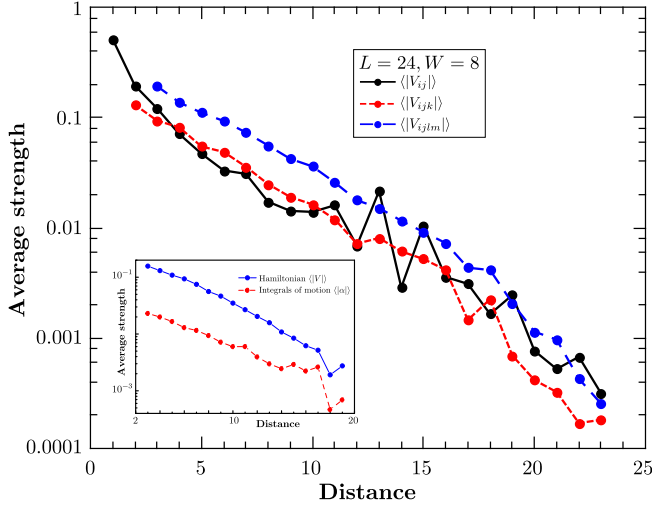


FIG. 1. The average of the absolute value of the coefficients $\langle |V_{i_1, \dots, i_n}| \rangle$ present in the classical Hamiltonian. Here we show results deep in the many-body localized phase with $W = 8$, for a system size $L = 24$ averaged over 20 disorder realizations. The different curves represent the two-body V_{ij} , three-body V_{ijk} , and four-body V_{ijkl} coefficients, respectively. The strength of the coupling constants is almost independent of the order of the interaction. Inset: The spread of the local integrals of motion at disorder strength $W = 8$ and $L = 20$, compared to the coefficients of the Hamiltonian at the same distance. Note the difference between the localization length of the integrals of motion and the exponential decay of the interactions.

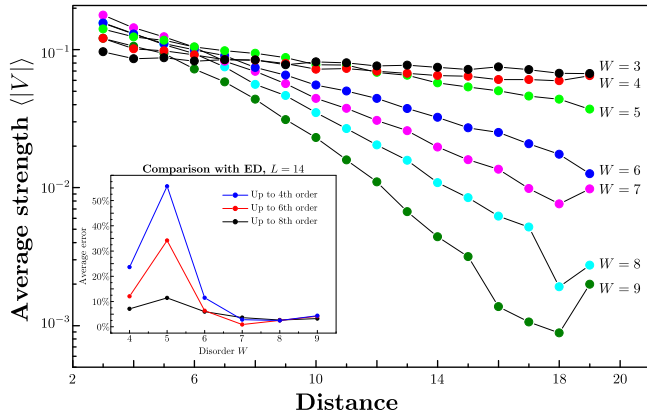


FIG. 2. The average of the absolute value of the coefficients $\langle |V_{i_1, \dots, i_n}| \rangle$ as a function of distance for various disorder strengths $W = 3, \dots, 9$ for $L = 20$. In the localized phase one can see exponential decay, whereas in the delocalized phase at $W < W_c \approx 5$ the coefficients are independent of distance. Inset: The relative error in the ground-state energy obtained using displacement transformations, compared to exact diagonalization results for $L = 14$. The different curves represent the cutoff at a given order $n = 4, 6, 8$. In the localized phase even few-body interactions are sufficient to reproduce the ED results. In the delocalized regime one needs to go up to order 8, whereas around the MBL transition interactions of even higher order need to be included.

$$\tau_i^z = U^\dagger n_i U = n_i + \sum_{jklm} \alpha_{jklm}^i c_j^\dagger c_k c_l^\dagger c_m + \dots, \quad (16)$$

where U is the product of all the displacement transformations. Note that at quadratic order $\tau_i^z = n_i$, consistent with the fact that the single-particle spectrum is unchanged by the interactions.

The average of the absolute value of the coefficients α as a function of the distance away from the original site i is shown in the inset of Fig. 1. The integrals of motion are indeed exponentially localized, as expected. The localization length for τ^z is, however, different from the length associated with the decay of the interactions coefficients V_{ij} .

These results serve as a proof of principle that the method of displacement transformations can be used to study interacting fermion models. We also applied our method to a different model in the Supplemental Material [13]. In future work, we will apply this method to extract new physical results for various models.

Outlook.—We introduced a sequence of displacement transformations that allows for the diagonalization of an interacting fermionic Hamiltonian. This method suggests we can bring *any* charge-conserving Hamiltonian into the classical form of Eq. (1), not limited to the many-body localized phase [11] or $d = 1$. For example, even the completely nonlocal Fermi liquids [15] can be analyzed using the classical model [16]

$$E = \sum_k \xi_k n_k + \frac{1}{2} \sum_{kk'} f_{kk'} n_k n_{k'} + \dots, \quad (17)$$

where the integrals of motion n_k are now local in momentum space and, thus, delocalized in real space. This begs the question how our method is related to the notion of integrability. We propose that in Eq. (1) quantum integrability depends on the number of parameters $V_{i_1, \dots, i_n}$ that are nonzero, following the definition of Ref. [17]: if the number of parameters that are nonzero and independent is subexponential or less, the system is integrable. Note that this case of sparse uncorrelated coupling constants also implies Poissonian-level statistics [18]. Our method thus most effectively applies to systems close to integrability. The interesting question of how ergodic systems are represented in the τ basis of Eq. (1) remains open.

A possible fruitful future endeavor would be to recast the iterative transformations in the language of an analytic renormalization scheme, much like the strong-disorder renormalization group theory [19,20], for which a related version has been constructed for MBL systems [21,22]. A similar method using a Hilbert space preserving renormalization group scheme has been introduced by You *et al.* [23]. The difference is that they keep interaction terms of all orders, and instead treat the off-diagonal resonance perturbatively to second order.

Even though the final classical Hamiltonian is of remarkable simplicity, it does not imply easy solutions, because in principle there could be long-range effective interactions between the classical bits τ^z . Furthermore, some complexity of the initial Hamiltonian is transferred to the transformation operator U , which is needed to translate any physical operator into the τ basis. The fact that we can explicitly derive U using the renormalization scheme described in this Letter, though, introduces a novel quantitative tool for the study of strongly interacting quantum matter.

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